

Qualification Pack



Compressed Biogas Plant Technician

QP Code: HYC/Q4402

Version: 1.0

NSQF Level: 4

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HYC/Q4402: Compressed Biogas Plant Technician

Brief Job Description

Compressed Biogas Plant technician will be responsible for maintaining the optimum plant conditions for biogas production, select proper size and design of biogas plant and to construct a biogas plant, perform testing of biogas plants identify and locate the problems in biogas plants and repair them etc.

Personal Attributes

The individual should have a technical background & aptitude, ability to give attention to detail, adaptability, teamwork, and a strong commitment & passion towards safety and sustainability. The person should have the ability to think creatively and work individually and in a team for extended hours in confined spaces or at height. Experience with operating basic site machinery (e.g., forklift/telehandler, boom lift, scissor lift) is an advantage.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N4403: Introduction to Compressed Biogas](#)
2. [HYC/N4404: CBG Plant and Machinery](#)
3. [HYC/N4405: Testing and Repairing in a CBG Plant](#)
4. [HYC/N9301: Working Effectively in a team](#)
5. [HYC/N9302: Maintain health, safety and security procedures](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Bio Fuel Production - Operation & Maintenance
Country	India
NSQF Level	4

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Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO2015/3134
Minimum Educational Qualification & Experience	12th grade Pass (Science) with 1-2 Years of experience OR Completed 3 year diploma after 10th (in relevant branch) OR 10th grade pass with 3 Years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	29/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-04-HY-02619-2024-V1-HSSC
NQR Version	1.0

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HYC/N4403: Introduction to Compressed Biogas

Description

This unit provides general introduction about the biogas plants in general and about compressed biogas (CBG) plants in particular, general process of producing the biogas, feedstocks/raw material for biogas production, anaerobic decomposition process, significance of optimum Condition for Biogas production, production of compressed biogas (CBG) from biogas and their applications.

Scope

The scope covers the following :

- This unit/task covers the following:
- Biogas/CBG production process
- Raw materials for biogas production
- Compressed biogas and applications

Elements and Performance Criteria

Introduction to Compressed Biogas

To be competent, the user/individual on the job must be able to:

- PC1.** Define what is biogas
- PC2.** Know its composition and energy content
- PC3.** Understand why it is also called Bio-CNG
- PC4.** Understand the basic process of anaerobic decomposition/digester
- PC5.** Identify different steps involved in the process like crushing, slurry preparation, anaerobic digestion (fermentation), Biogas Purification, Bio-NG Compression & Cylinder or Cascade Gas Filling
- PC6.** Identify key factors affecting anaerobic biodigester performance
- PC7.** Appreciate importance of different micro-organisms involved in biogas production
- PC8.** Know the role of methanogenic microorganisms
- PC9.** Identify different raw materials which can be used to produce biogas
- PC10.** Identify by-products produced during the process
- PC11.** Understand wastes being generated and its handling/disposal
- PC12.** Know difference between biogas and compressed biogas
- PC13.** Define benefits of biogas and CBG
- PC14.** Know potential of CBG in India
- PC15.** Appreciate importance of raw material availability for CBG production
- PC16.** Know about different types of bio-digesters and plant configuration
- PC17.** Identify and understand different units/components in a CBG plant
- PC18.** Understand the laboratory testing and data recording
- PC19.** Know the proper sampling procedures

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PC20. Know applications of biogas and CBG

PC21. Understand safety issues with CBG

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the general process for producing biogas/CBG
- KU2.** Principles of anaerobic digestion process and role of microorganisms
- KU3.** What all raw materials can be used to produce biogas/CBG
- KU4.** How composition of raw material will affect the biogas production
- KU5.** Different types of anaerobic bio-digester being employed
- KU6.** Benefits from using CBG w.r.t to environment and energy security
- KU7.** Potential of CBG in India w.r.t biomass availability
- KU8.** Units / components in a CBG plant
- KU9.** Safety related precautions to be taken with CBG
- KU10.** Different applications of biogas in general and CBG in particular
- KU11.** communication skills for conveying equipment and work requirements
- KU12.** reporting protocol and documentation required
- KU13.** the importance of personal protective equipments
- KU14.** Importance of maintaining all the units in working conditions
- KU15.** Safety protocols in handling CBG
- KU16.** The concept of Mission Karmayogi
- KU17.** The concept of Swachata hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret the standard operating procedures for Compressed Biogas work
- GS2.** effective use of computers (MS office tools)
- GS3.** presentation skills -oral and written.
- GS4.** read instructions, guidelines/procedures and reports
- GS5.** record data on waste disposal at workplace
- GS6.** assess complexity and criticality of task
- GS7.** communicate effectively with team members
- GS8.** perform basic mathematical calculation
- GS9.** analyze the parameters and output
- GS10.** complete tasks efficiently and accurately within stipulated time
- GS11.** work constructively and collaboratively with others.
- GS12.** plan and organize the activities

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Compressed Biogas</i>	25	45	-	-
PC1. Define what is biogas	1	2	-	-
PC2. Know its composition and energy content	1	2	-	-
PC3. Understand why it is also called Bio-CNG	1	2	-	-
PC4. Understand the basic process of anaerobic decomposition/digester	1	2	-	-
PC5. Identify different steps involved in the process like crushing, slurry preparation, anaerobic digestion (fermentation), Biogas Purification, Bio-NG Compression & Cylinder or Cascade Gas Filling	2	3	-	-
PC6. Identify key factors affecting anaerobic biodigester performance	2	2	-	-
PC7. Appreciate importance of different micro-organisms involved in biogas production	1	2	-	-
PC8. Know the role of methanogenic microorganisms	1	2	-	-
PC9. Identify different raw materials which can be used to produce biogas	1	2	-	-
PC10. Identify by-products produced during the process	1	2	-	-
PC11. Understand wastes being generated and its handling/disposal	1	2	-	-
PC12. Know difference between biogas and compressed biogas	2	2	-	-
PC13. Define benefits of biogas and CBG	1	2	-	-
PC14. Know potential of CBG in India	1	2	-	-
PC15. Appreciate importance of raw material availability for CBG production	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Know about different types of bio-digesters and plant configuration	2	3	-	-
PC17. Identify and understand different units/components in a CBG plant	1	2	-	-
PC18. Understand the laboratory testing and data recording	1	3	-	-
PC19. Know the proper sampling procedures	1	2	-	-
PC20. Know applications of biogas and CBG	1	2	-	-
PC21. Understand safety issues with CBG	1	2	-	-
NOS Total	25	45	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4403
NOS Name	Introduction to Compressed Biogas
Sector	Hydrocarbon
Sub-Sector	
Occupation	Bio Fuel Production - Operation & Maintenance
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

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HYC/N4404: CBG Plant and Machinery

Description

This unit covers details about different types of biogas plants, their configuration, different mechanical systems involved and installation and commissioning activities.

Scope

The scope covers the following :

- This unit/task covers the following:
- Plant & machinery
- Types of biogas plants
- Fundamentals of mechanical systems in a digester

Elements and Performance Criteria

CBG Plant and Machinery

To be competent, the user/individual on the job must be able to:

- PC1.** Know different components of a biogas plant
- PC2.** Understand main anaerobic digester working
- PC3.** Identify the role of slurry storage tank
- PC4.** Know the type of slurry and sludge pumps used
- PC5.** Understand types of organic waste crusher shredder machine
- PC6.** Know different types /configuration of anaerobic digesters
- PC7.** Identify classification of biogas plants depending upon plant design and mode of working
- PC8.** Know about Batch Type Biogas Plant and Continuous Type biogas Plant
- PC9.** Know about movable Type Drums biogas plant
- PC10.** Understand fundamental of mechanical systems in a biogas plant
- PC11.** Identify which type of pumps to be used for a type of feed stock/raw material, Total solid (TS) concentration
- PC12.** Know the importance of system redundancy
- PC13.** Understand piping system design based on flow rates, pressure, Moisture content, Temperature etc
- PC14.** Understand importance of regular mixing in a digester
- PC15.** Know different types of mixers to be used
- PC16.** Know other important components like feedstock conveyer
- PC17.** Understand importance of biogas leak detection systems
- PC18.** Know the working gas detection and alarm systems to ensure safety
- PC19.** Identify pressure regulators and other valves in the plant
- PC20.** Know about biogas blowers and compressors

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different parts of a biogas plant
- KU2.** Working of main digester
- KU3.** Choice and working different slurry/sludge pumps
- KU4.** Why shredder and crushing of feed stock is important
- KU5.** The biogas plant type and configuration
- KU6.** Which pumps to be used based on solid content
- KU7.** Why system redundancy is important in a biogas plant
- KU8.** Working of other minor and major components in a CBG plant like piping systems, mixer, valves etc
- KU9.** Gas compression for making CBG
- KU10.** Working of gas leakage, detection and alarm systems
- KU11.** communication skills for conveying equipment and work requirements
- KU12.** reporting protocol and documentation required
- KU13.** the importance of personal protective equipments
- KU14.** Importance of maintaining all the units in working conditions
- KU15.** Safety protocols in handling CBG
- KU16.** The concept of Mission Karmayogi
- KU17.** The concept of Swachhata hi Seva.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret the standard operating procedures for Compressed Biogas work
- GS2.** effective use of computers (MS office tools)
- GS3.** presentation skills -oral and written.
- GS4.** read instructions, guidelines/procedures and reports
- GS5.** record data on waste disposal at workplace
- GS6.** assess complexity and criticality of task
- GS7.** communicate effectively with team members
- GS8.** perform basic mathematical calculation
- GS9.** analyze the parameters and output
- GS10.** complete tasks efficiently and accurately within stipulated time
- GS11.** work constructively and collaboratively with others.
- GS12.** plan and organize the activities

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>CBG Plant and Machinery</i>	20	40	-	-
PC1. Know different components of a biogas plant	1	2	-	-
PC2. Understand main anaerobic digester working	1	2	-	-
PC3. Identify the role of slurry storage tank	1	2	-	-
PC4. Know the type of slurry and sludge pumps used	1	2	-	-
PC5. Understand types of organic waste crusher shredder machine	1	2	-	-
PC6. Know different types /configuration of anaerobic digesters	1	2	-	-
PC7. Identify classification of biogas plants depending upon plant design and mode of working	1	2	-	-
PC8. Know about Batch Type Biogas Plant and Continuous Type biogas Plant	1	2	-	-
PC9. Know about movable Type Drums biogas plant	1	2	-	-
PC10. Understand fundamental of mechanical systems in a biogas plant	1	2	-	-
PC11. Identify which type of pumps to be used for a type of feed stock/raw material, Total solid (TS) concentration	1	2	-	-
PC12. Know the importance of system redundancy	1	2	-	-
PC13. Understand piping system design based on flow rates, pressure, Moisture content, Temperature etc	1	2	-	-
PC14. Understand importance of regular mixing in a digester	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Know different types of mixers to be used	1	2	-	-
PC16. Know other important components like feedstock conveyer	1	2	-	-
PC17. Understand importance of biogas leak detection systems	1	2	-	-
PC18. Know the working gas detection and alarm systems to ensure safety	1	2	-	-
PC19. Identify pressure regulators and other valves in the plant	1	2	-	-
PC20. Know about biogas blowers and compressors	1	2	-	-
NOS Total	20	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4404
NOS Name	CBG Plant and Machinery
Sector	Hydrocarbon
Sub-Sector	
Occupation	Bio Fuel Production - Operation & Maintenance
NSQF Level	4
Credits	3.5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

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HYC/N4405: Testing and Repairing in a CBG Plant

Description

This unit covers details of various testing related requirements as well as maintenance & repairing activities in a CBG plant.

Scope

The scope covers the following :

- This unit/task covers the following:
- Laboratory Testing
- System inspection and maintenance

Elements and Performance Criteria

Laboratory Testing

To be competent, the user/individual on the job must be able to:

- PC1.** Identify what different routine testing is to be done in a CBG plant
- PC2.** Know significance of regular calibration
- PC3.** Understand how to measure Total Solids (TS) in a slurry
- PC4.** Test volatile solids (VS) in TS
- PC5.** Test pH in a digester
- PC6.** Test other parameters like total alkalinity and temperature etc
- PC7.** Measure methane and CO₂ content in biogas
- PC8.** Monitor other important parameters like COD, BOD, nutrients, metals, Nitrogen, Phosphorous, Potassium etc
- PC9.** Know the working of different online analysers and laboratory equipments
- PC10.** Follow proper sampling procedure and frequency of testing as per manufacturer protocols and SOPs
- PC11.** Report the data as per prescribed format to the operator

System inspection and maintenance

To be competent, the user/individual on the job must be able to:

- PC12.** Appreciate importance of system inspection & maintenance
- PC13.** Identify routine versus major maintenance
- PC14.** Know the Impact of Process Conditions on Frequency of maintenance
- PC15.** Check digester tanks periodically for leaks, cracks, or other evidence of structural problems
- PC16.** periodically check and operate pressure relief valves (PRVs)
- PC17.** Check the piping and equipment for leaks daily
- PC18.** Monitor the corrosion on a regular basis
- PC19.** Replace any Worn or damaged seals and/or drive belt in a pump
- PC20.** Check the mechanical mixers for any excessive vibration and rectify the same

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- PC21.** Identify any fouling of the mixer blades and clear any deposited scale/cake etc
- PC22.** Monitor the working of compressor and change the oil as and when required
- PC23.** Check and change the particulate filter, Coalescing filters and O-rings etc
- PC24.** Perform regular tasks like checking for leaks, verifying proper calibration and operation of the instrumentation, biogas analysers in a H₂S removal system
- PC25.** check coolant temperature, glycol-water solution level, and drive belts etc in the moisture removal system
- PC26.** Replace the membrane filter cartridges in membrane-based CO₂ removal system or the adsorbents in a pressure swing adsorption (PSA) based CO₂ removal system
- PC27.** Ensure proper working of gas detection system to prevent any accident

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Testing requirements in a biogas plant
- KU2.** Operation and maintenance of various testing equipments
- KU3.** Working of different online analyzers and their maintenance
- KU4.** Handling of various general tools and kits required for inspection and maintenance in a CBG plant
- KU5.** How to repair replace faulty parts or components in different units
- KU6.** Importance of maintaining inventory of spare parts, lubricating oil, filters, seals etc
- KU7.** Working of gas leakage, detection and alarm systems
- KU8.** communication skills for conveying equipment and work requirements
- KU9.** reporting protocol and documentation required
- KU10.** the importance of personal protective equipments
- KU11.** Importance of maintaining all the units in working conditions
- KU12.** Safety protocols in handling CBG
- KU13.** The concept of Mission karmayogi
- KU14.** The concept of Swachhata hi Seva.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret the standard operating procedures for Compressed Biogas work
- GS2.** effective use of computers (MS office tools)
- GS3.** presentation skills -oral and written.
- GS4.** read instructions, guidelines/procedures and reports
- GS5.** record data on waste disposal at workplace
- GS6.** assess complexity and criticality of task
- GS7.** communicate effectively with team members

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- GS8.** perform basic mathematical calculation
- GS9.** analyze the parameters and output
- GS10.** complete tasks efficiently and accurately within stipulated time
- GS11.** work constructively and collaboratively with others.
- GS12.** plan and organize the activities

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Laboratory Testing</i>	12	22	-	-
PC1. Identify what different routine testing is to be done in a CBG plant	2	2	-	-
PC2. Know significance of regular calibration	1	2	-	-
PC3. Understand how to measure Total Solids (TS) in a slurry	1	2	-	-
PC4. Test volatile solids (VS) in TS	1	2	-	-
PC5. Test pH in a digester	1	2	-	-
PC6. Test other parameters like total alkalinity and temperature etc	1	2	-	-
PC7. Measure methane and CO ₂ content in biogas	1	2	-	-
PC8. Monitor other important parameters like COD, BOD, nutrients, metals, Nitrogen, Phosphorous, Potassium etc	1	2	-	-
PC9. Know the working of different online analysers and laboratory equipments	1	2	-	-
PC10. Follow proper sampling procedure and frequency of testing as per manufacturer protocols and SOPs	1	2	-	-
PC11. Report the data as per prescribed format to the operator	1	2	-	-
<i>System inspection and maintenance</i>	18	33	-	-
PC12. Appreciate importance of system inspection & maintenance	1	2	-	-
PC13. Identify routine versus major maintenance	2	2	-	-
PC14. Know the Impact of Process Conditions on Frequency of maintenance	1	2	-	-
PC15. Check digester tanks periodically for leaks, cracks, or other evidence of structural problems	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. periodically check and operate pressure relief valves (PRVs)	1	2	-	-
PC17. Check the piping and equipment for leaks daily	1	2	-	-
PC18. Monitor the corrosion on a regular basis	1	2	-	-
PC19. Replace any Worn or damaged seals and/or drive belt in a pump	1	2	-	-
PC20. Check the mechanical mixers for any excessive vibration and rectify the same	1	2	-	-
PC21. Identify any fouling of the mixer blades and clear any deposited scale/cake etc	2	2	-	-
PC22. Monitor the working of compressor and change the oil as and when required	1	2	-	-
PC23. Check and change the particulate filter, Coalescing filters and O-rings etc	1	2	-	-
PC24. Perform regular tasks like checking for leaks, verifying proper calibration and operation of the instrumentation, biogas analysers in a H ₂ S removal system	1	3	-	-
PC25. check coolant temperature, glycol-water solution level, and drive belts etc in the moisture removal system	1	2	-	-
PC26. Replace the membrane filter cartridges in membrane-based CO ₂ removal system or the adsorbents in a pressure swing adsorption (PSA) based CO ₂ removal system	1	2	-	-
PC27. Ensure proper working of gas detection system to prevent any accident	1	2	-	-
NOS Total	30	55	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4405
NOS Name	Testing and Repairing in a CBG Plant
Sector	Hydrocarbon
Sub-Sector	
Occupation	Bio Fuel Production - Operation & Maintenance
NSQF Level	4
Credits	3.5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

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HYC/N9301: Working Effectively in a team

Description

This unit is about working effectively within a team.

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** provide support to the team members
- PC4.** respect the colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team
- PC8.** adopt gender neutral behaviour while interacting with colleagues
- PC9.** offer assistance to a person with disability (PWD), only if required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** the importance of creating an environment of trust and mutual respect
- KU4.** the implications of own work on the work and schedule of others
- KU5.** the standard practices in organisation w.r.t communication at various levels
- KU6.** the personal responsibility for completing the task in time
- KU7.** importance of gender equality
- KU8.** importance of showing empathy while interacting with a PwD

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** work in a disciplined manner for meeting commitments and deadline
- GS4.** how to plan and prioritise the work
- GS5.** the importance of consistent and reliable services
- GS6.** apply problem solving approaches in different situations
- GS7.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	3	-	-
PC2. pass on information to colleagues in line with organisational requirements	2	3	-	-
PC3. provide support to the team members	2	4	-	-
PC4. respect the colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	2	3	-	-
PC6. inform team members timely, if timelines can't be met	2	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
PC8. adopt gender neutral behaviour while interacting with colleagues	2	2	-	-
PC9. offer assistance to a person with disability (PWD), only if required	2	3	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working Effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

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HYC/N9302: Maintain health, safety and security procedures

Description

This unit is about maintaining health, safety and security procedure at workplace. It covers responsibilities towards self, others, assets and the environment.

Scope

The scope covers the following :

- Follow health and safety measures
- Follow safety procedures during emergency

Elements and Performance Criteria

Follow health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions
- PC2.** identify the people responsible for maintaining health and safety in the workplace
- PC3.** identify possible causes of risk or accident in the workplace
- PC4.** follow safe working practices while dealing with hazards to ensure the safety of self and others
- PC5.** lift heavy objects safely using correct procedures
- PC6.** follow safety signages
- PC7.** maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub
- PC8.** inform the concerned person of any illness related to self and others
- PC9.** maintain workplace hygiene by disinfecting the equipment and tools regularly

Follow safety procedures during emergency

To be competent, the user/individual on the job must be able to:

- PC10.** respond promptly and appropriately to an accident or in an emergency situation
- PC11.** use appropriate fire extinguishers for different types of fires correctly
- PC12.** follow appropriate rescue techniques during fire hazard
- PC13.** follow good housekeeping practice in order to prevent fire hazards
- PC14.** inform fire safety department about any near-miss incidents in the work place
- PC15.** provide appropriate first aid to victims in an emergency situation
- PC16.** follow the applicable regulations and codes as per safety standard
- PC17.** prepare written accident/incident report and share with the concerned officer/department

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** company's policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** health and safety hazards commonly affecting the work environment and related precautions
- KU4.** importance of maintaining personal hygiene using PPE kit, sanitizer and soap
- KU5.** importance of maintaining workplace hygiene
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** identify and report potential sources of danger
- GS4.** how to plan the work to meet the deadline
- GS5.** the importance of on time services
- GS6.** apply problem solving approaches in different situations
- GS7.** apply balanced judgments in different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and safety measures</i>	9	15	-	-
PC1. use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions	1	2	-	-
PC2. identify the people responsible for maintaining health and safety in the workplace	1	-	-	-
PC3. identify possible causes of risk or accident in the workplace	1	2	-	-
PC4. follow safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC5. lift heavy objects safely using correct procedures	1	2	-	-
PC6. follow safety signages	1	2	-	-
PC7. maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub	1	2	-	-
PC8. inform the concerned person of any illness related to self and others	1	1	-	-
PC9. maintain workplace hygiene by disinfecting the equipment and tools regularly	1	2	-	-
<i>Follow safety procedures during emergency</i>	11	15	-	-
PC10. respond promptly and appropriately to an accident or in an emergency situation	1	2	-	-
PC11. use appropriate fire extinguishers for different types of fires correctly	2	2	-	-
PC12. follow appropriate rescue techniques during fire hazard	1	2	-	-
PC13. follow good housekeeping practice in order to prevent fire hazards	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inform fire safety department about any near-miss incidents in the work place	2	2	-	-
PC15. provide appropriate first aid to victims in an emergency situation	1	2	-	-
PC16. follow the applicable regulations and codes as per safety standard	1	2	-	-
PC17. prepare written accident/incident report and share with the concerned officer/department	2	2	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9302
NOS Name	Maintain health, safety and security procedures
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N4403.Introduction to Compressed Biogas	25	45	-	-	70	20
HYC/N4404.CBG Plant and Machinery	20	40	-	-	60	20
HYC/N4405.Testing and Repairing in a CBG Plant	30	55	-	-	85	20
HYC/N9301.Working Effectively in a team	20	30	-	-	50	10
HYC/N9302.Maintain health, safety and security procedures	20	30	-	-	50	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	135	230	-	-	365	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
Gs	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
SS	Stainless Steel

Qualification Pack

PPE

Personal Protective Equipment

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.